DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B. Tech **Branch: AIDS/CSE(AI&ML)/CSE(AI)/CSE (DS)** **Semester: V**

Subject Code & Name: **BTAIC502 Machine Learning**

Max Marks: 60 **Date:08/02/2025** **Duration: 3 Hr.**

Instructions to the Students:

1. *Each question carries 12 marks.*
2. *Question No. 1 will be compulsory and include objective-type questions.*
3. *Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.*
4. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
5. *Use of non-programmable scientific calculators is allowed.*
6. *Assume suitable data wherever necessary and mention it clearly.*

					(Level/CO)	Marks
Q. 1	**Objective type questions. (Compulsory Question)**					**12**
1	Which of the following is **not** a category of Machine Learning?				Remember CO1	**1**
	a. Supervised Learning	b. unsupervised Learning	c.reinforcement Learning	d. Manual Learning		
2	Which of the following is **not** an application of Machine Learning?				Remember CO1	**1**
	a. Spam email detection	b. Image recognition	c. Word processing software	d. Fraud detection		
3	Which dataset is used to evaluate the final performance of a trained model?				Remember CO1	**1**
	a. Training data	b. Validation data	c. Testing data	d. Preprocessing data		
4	In a confusion matrix, which term represents correctly predicted positive cases?				Understand CO2	**1**
	a. True Negatives (TN)	b. True Positives (TP)	c. False Positives (FP)	d. False Negatives (FN)		
5	An AUC score of 1.0 indicates:				Understand CO2	**1**
	a. Perfect classification	b. Random guessing	c. Poor model performance	d. Overfitting		
6	The cost function in Linear Regression is typically:				Understand CO3	**1**
	a. Mean Squared Error (MSE)	b. Logarithmic Loss	c. Cross-Entropy Loss	d. Mean Absolute Error (MAE)		
7	Multivariable regression differs from simple linear regression in that:				Understand CO3	**1**
	a. It can handle more than one independent variable	b. It uses a different cost function	c. It cannot use gradient descent	d. It only works with categorical variables		

8	In a Decision Tree, the process of deciding the feature to split on is based on: a. Feature correlation b. Information Gain, Gain Ratio, or Gini Index c. Random feature selection d. The size of the dataset	Remember CO4	**1**
9	Random Forests can be used for: a. Classification problems only b. Regression problems only c. Both classification and regression problems d. Clustering problems	Remember CO4	**1**
10	Naive Bayes can be effectively used for: a. Regression problems b. Clustering problems c. Text classification problems d. Feature scaling	Remember CO5	**1**
11	Which library in Python is commonly used for implementing KNN? a. TensorFlow b. Pandas c. Sklearn (scikit-learn) d. NumPy	Remember CO5	**1**
12	Which function is used in SVM to find a non-linear decision boundary? a. Entropy function b. Sigmoid function c. Kernel function d. Gradient function	Remember CO5	**1**
Q. 2	**Solve the following.**		**12**
A)	Define Machine Learning and the concept of learning in Machine Learning. Compare Machine Learning with traditional programming approaches.	Remember CO1	**6**
B)	What is the purpose of splitting data into training, validation, and testing datasets in Machine Learning? Explain each dataset's role with an example	Understand CO1	**6**
Q.3	**Solve the following.**		**12**
A)	A binary classification model's confusion matrix is given as follows: **Predicted: Positive Negative** **Actual Positive 70 30** **Actual Negative 20 80** Calculate the precision, recall (sensitivity), F1 score, and accuracy for this model. Explain which metric would be most important if this were a medical diagnostic test.	Apply CO2	**6**
B)	Explain the difference between Mean Absolute Error (MAE) and Mean Squared Error (MSE). Why might one metric be preferred over the other in specific scenarios?	Understand CO2	**6**

Q. 4	**Solve Any Two of the following.**		**12**
A)	Explain the intuition behind linear regression. How are the optimal coefficients determined, and what role does the cost function play in this process?	Understand CO3	**6**
B)	What is gradient descent, and how does it help in finding the optimal coefficients for a multivariable regression model? Discuss the impact of the learning rate on the convergence of the gradient descent algorithm.	Remember CO3	**6**
C)	Compare and contrast linear regression and logistic regression in terms of their applications, cost functions, and model assumptions.	Analyze CO3	**6**
Q.5	**Solve Any Two of the following.**		**12**
A)	Explain the process of building a decision tree. How does the algorithm decide which feature to split on at each node? What criteria are used to evaluate which feature provides the best split?	Understand CO4	**6**
B)	What are Extra Trees, and how do they differ from standard decision trees in a random forest? Discuss the advantages of using Extra Trees over regular decision trees.	Remember CO4	**6**
C)	How can decision trees and random forests be used for regression tasks? What are the key differences compared to classification?	Analyze CO4	**6**
Q. 6	**Solve Any Two of the following.**		**12**
A)	Explain Bayes' Theorem and how it is applied in Naive Bayes classification. Discuss the role of the independence assumption in simplifying the calculation of conditional probabilities and its implications on the model's performance.	Understand CO5	**6**
B)	How do you determine the optimal value of K for a K-Nearest Neighbors classifier? Discuss the effect of different values of K on model performance, and how cross-validation can help in selecting the best K.	Analyze CO5	**6**
C)	Explain the concept of landmark points in Support Vector Machines and their role in defining the decision boundary. How do similarity functions (e.g., linear, polynomial, Gaussian) impact the decision boundary and model performance?	Remember CO5	**6**
	*** **End** ***		